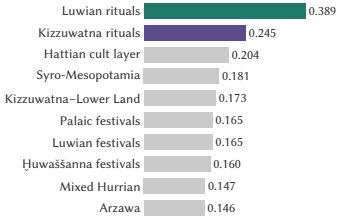
IN WORDS	ON THE FRAGMENT	AS A FORMULA
1 Read. An orphan, KBo 41.22, filed under CTH 470. Each legible word gets a word class from the edition's analysis; dashed: unclear.	3′ i-da-a-lu-un UD-an ma- 4′ DINGIR^{MEŠ}-aš kar-pí-in pa-an-ga-u-wa-aš EME-an 5′ ša-me-ed-du-wa-ra-at-kán ḫi-x 6′ na-aš-ta EN SÍSKUR ŠUM-ŠU 7′ le-e ku-wa-pí-ik-ki a-x 8′ te-eš-ḫa-an ma-ni-ku-wa-an(-) 9′ nu-uš-ši ^{GI}pát-tar pa-ra-a 10′ ^{MUNUS}ŠU.GI-ma me-ma-i x 11′ DINGIR^{MEŠ}-aš kar-pí-in pa-an-ga-u-wa-aš 12′ ^{LÚ}A.ZU-ma-aš-ši-iš-ša-an 13′ na-aš ša-ra-a ti- 14′ na-aš-ta GU$_4$-un 15′ a-ra-aḫ-za-an-da function verb noun, realia agent, deity foreign unclear	33 legible words.
2 Count. Every word form and every sign sequence inside a word is a feature. A feature counts more when it is frequent in the fragment and rare in the corpus.	pa-an-ga-u-wa-aš → **pankauaš** 0.18, *pa-an-ga* 0.16, *an-ga-u* 0.16, *ga-u-wa* 0.17 Rare word **pankauaš**: idf 6.8. Common sign *na*: idf 1.2. The fragment becomes a vector $\mathbf{x}$ with 177 non-zero entries out of 25,372.	$x_f \propto (1+\log tf_f)\, idf_f$ $idf_f = \log \frac{1+N}{1+df_f} + 1$ $\lVert \mathbf{x} \rVert = 1$
3 Compare. Each tradition is the average of its manuscripts. The fragment is compared with each average.		$\mathbf{c}_t = \frac{\sum_{d \in t} \mathbf{x}_d}{\lVert \sum_{d \in t} \mathbf{x}_d \rVert}$ $s_t = \cos(\mathbf{x}, \mathbf{c}_t) = \sum_f x_f c_{t,f}$
4 Decide. Place the fragment only if the nearest tradition leads the runner-up clearly; otherwise leave it unassigned.	$m = 0.14 \geq 0.018 \Rightarrow$ **Luwian rituals**	$m = s_1 - s_2 = 0.39 - 0.25$ place if $m \geq \tau = 0.018$ (τ: 80% of test fragments right)
5 Explain. The lead is a sum over features, so each word's share of the decision can be shown. Here the formula *pa-an-ga-u-wa-aš EME-an* decides.	3′ i-da-a-lu-un UD-an ma- 4′ DINGIR^{MEŠ}-aš kar-pí-in pa-an-ga-u-wa-aš EME-an 5′ ša-me-ed-du-wa-ra-at-kán ḫi-x 6′ na-aš-ta EN SÍSKUR ŠUM-ŠU 7′ le-e ku-wa-pí-ik-ki a-x 8′ te-eš-ḫa-an ma-ni-ku-wa-an(-) 9′ nu-uš-ši ^{GI}pát-tar pa-ra-a 10′ ^{MUNUS}ŠU.GI-ma me-ma-i x 11′ DINGIR^{MEŠ}-aš kar-pí-in pa-an-ga-u-wa-aš 12′ ^{LÚ}A.ZU-ma-aš-ši-iš-ša-an 13′ na-aš ša-ra-a ti- 14′ na-aš-ta GU$_4$-un 15′ a-ra-aḫ-za-an-da Darker: larger share of the lead.	$m = \sum_f \underbrace{x_f (c_{1,f} - c_{2,f})}_{\text{share of feature } f}$
6 Check by hand. The closest single manuscripts are candidates for the philologist.	The five nearest manuscripts are all Tunnawiya's ritual (CTH 409), which no reference group includes.	$\arg\max_d \cos(\mathbf{x}, \mathbf{x}_d)$